

Assignment – 1

Subject: Advanced Java Programming

Unit: Java Networking Fundamentals

Branch: B.E. ICT (5th Semester)

Dhwanit Patel

240170132044

Part C - Programming

1. Display the local host name and IP address using InetAddress.

```
import java.net.InetAddress;

class Q11 {
    public static void main(String[] args) throws Exception {
        InetAddress localhost = InetAddress.getLocalHost();
        System.out.println("Host Name: " + localhost.getHostName());
        System.out.println("IP Address: " + localhost.getHostAddress());
    }
}
```

```
● PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\workspace\.metadata\.plugins\org.eclipse.jdt_ws\PBL_1_43a4a77f\bin' 'Q11'
Host Name: DESKTOP-3VPSEA8
IP Address: 192.168.31.22
○ PS D:\Sem-5\AJP\PBL_1> █
```

2. Display the IP address of a given website using InetAddress.

```
import java.net.InetAddress;
import java.util.Scanner;

class Q12 {
    public static void main(String[] args) throws Exception {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter website name: ");
        String website = scanner.nextLine();
        InetAddress address = InetAddress.getByName(website);
        System.out.println("IP Address: " + address.getHostAddress());
    }
}
```

```

● PS D:\Sem-5\AJP\PBL_1> d:; cd 'd:\Sem-5\AJP\PBL_1'; &
howCodeDetailsInExceptionMessages' '-cp' 'C:\Users\BAPS
27c2fddc87b1f215e91539285\redhat.java\jdt_ws\PBL_1_43a4
Enter website name: youtube.com
IP Address: 142.250.134.91
○ PS D:\Sem-5\AJP\PBL_1>

```

- Using URL class, display protocol, host name, port number, file name and default port.

```

import java.net.URL;
import java.util.Scanner;

class Q13 {
    public static void main(String[] args) throws Exception {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter URL: ");
        URL url = new URL(scanner.nextLine());
        System.out.println("Protocol: " + url.getProtocol());
        System.out.println("Host Name: " + url.getHost());
        System.out.println("Port Number: " + url.getPort());
        System.out.println("File Name: " + url.getFile());
        System.out.println("Default Port: " + url.getDefaultPort());
    }
}

```

```

● PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin
ages' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\works
hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q13'
Enter URL: https://web.whatsapp.com/
Protocol: https
Host Name: web.whatsapp.com
Port Number: -1
File Name: /
Default Port: 443
○ PS D:\Sem-5\AJP\PBL_1>

```

- Using URLConnection, display content type, content length, last modified date and date of connection.

```

import java.net.URL;
import java.net.URLConnection;
import java.util.Date;
import java.util.Scanner;

class Q14 {
    public static void main(String[] args) throws Exception {

```

```

Scanner scanner = new Scanner(System.in);
System.out.print("Enter URL: ");
URL url = new URL(scanner.nextLine());
URLConnection connection = url.openConnection();
System.out.println("Content Type: " + connection.getContentType());
System.out.println("Content Length: " + connection.getContentLengthLong());
System.out.println("Last Modified: " + new Date(connection.getLastModified()));
System.out.println("Date of Connection: " + new Date(connection.getDate()));
}
}

```

```

● PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin\jav
ages' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspace
hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q14'
Enter URL: https://www.google.com/
Content Type: text/html; charset=ISO-8859-1
Content Length: -1
Last Modified: Thu Jan 01 05:30:00 IST 1970
Date of Connection: Sun Aug 02 17:17:47 IST 2026
○ PS D:\Sem-5\AJP\PBL_1>

```

5. Read data from a file using InputStream.

```

import java.io.FileInputStream;
import java.util.Scanner;

class Q15 {
    public static void main(String[] args) throws Exception {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter file path: ");
        FileInputStream fileInputStream = new FileInputStream(scanner.nextLine());
        int data;
        while ((data = fileInputStream.read()) != -1) {
            System.out.print((char) data);
        }
        fileInputStream.close();
    }
}

```

```

● PS D:\Sem-5\AJP\PBL_1> d:; cd 'd:\Sem-5\AJP\PBL_1'; & 'C:\Program Fil
howCodeDetailsInExceptionMessages' '-cp' 'C:\Users\BAPS\AppData\Roamin
27c2fddc87b1f215e91539285\redhat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q15'
Enter file path: D:\Sem-5\AJP\PBL_1\practicals\data.txt
hello this is dhwanit
I am from VGEC ICT
○ PS D:\Sem-5\AJP\PBL_1>

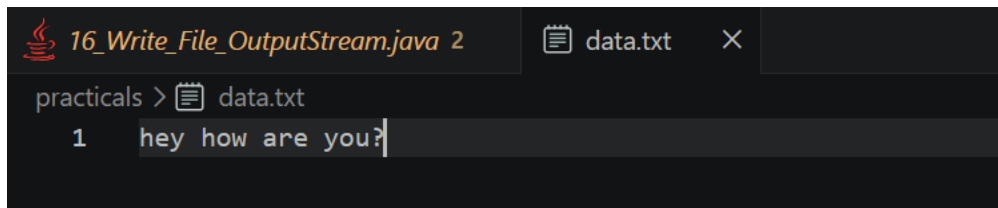
```

6. Write data to a file using OutputStream.

```
import java.io.FileOutputStream;
import java.util.Scanner;

class Q16 {
    public static void main(String[] args) throws Exception {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter file path: ");
        String filePath = scanner.nextLine();
        System.out.print("Enter data: ");
        String data = scanner.nextLine();
        FileOutputStream fileOutputStream = new FileOutputStream(filePath);
        fileOutputStream.write(data.getBytes());
        fileOutputStream.close();
    }
}
```

```
PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin\java.
ages' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspaceSto
hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q16'
Enter file path: D:\Sem-5\AJP\PBL_1\practicals\data.txt
Enter data: hey how are you?
PS D:\Sem-5\AJP\PBL_1>
```



16_Write_File_OutputStream.java 2 data.txt X

practicals > data.txt

1 hey how are you?

7. Write a TCP Client program to send a message to the Server.

```
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.PrintWriter;
import java.net.Socket;
import java.util.Scanner;

class Q17TCPClient {
    public static void main(String[] args) throws Exception {
        Socket socket = new Socket("localhost", 5000);
        BufferedReader input = new BufferedReader(new
        InputStreamReader(socket.getInputStream()));
        PrintWriter output = new PrintWriter(socket.getOutputStream(), true);
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter message: ");
```

```

        String message = scanner.nextLine();
        output.println(message);
        System.out.println("Server: " + input.readLine());
        socket.close();
    }
}

```

```

● PS D:\Sem-5\AJP\PBL_1> d:; cd 'd:\Sem-5\AJP\PBL_1'; & 'C:\Program Files\Java\jdk-25\bin\java.exe' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspaceStorage\27c2fddc87b1f215e91539285\redhat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q17'
Enter message: hello this is advance java practicals
Server: Message received
○ PS D:\Sem-5\AJP\PBL_1> 

```

8. Write a TCP Server program to receive a message from the Client.

```

import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.PrintWriter;
import java.net.ServerSocket;
import java.net.Socket;

class Q18TCPServer {
    public static void main(String[] args) throws Exception {
        ServerSocket serverSocket = new ServerSocket(5000);
        Socket socket = serverSocket.accept();
        BufferedReader input = new BufferedReader(new
        InputStreamReader(socket.getInputStream()));
        PrintWriter output = new PrintWriter(socket.getOutputStream(), true);
        String message = input.readLine();
        System.out.println("Client: " + message);
        output.println("Message received");
        socket.close();
        serverSocket.close();
    }
}

```

```

● PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin\java.exe' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspaceStorage\27c2fddc87b1f215e91539285\redhat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q18TCPServer'
Client: hello this is advance java practicals
○ PS D:\Sem-5\AJP\PBL_1> 

```

9. Develop a TCP Client-Server application to perform addition of two numbers.

```

import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.PrintWriter;
import java.net.Socket;
import java.util.Scanner;

class Q19AdditionClient {
    public static void main(String[] args) throws Exception {
        Socket socket = new Socket("localhost", 5001);
        BufferedReader input = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
        PrintWriter output = new PrintWriter(socket.getOutputStream(), true);
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter first number: ");
        output.println(scanner.nextLine());
        System.out.print("Enter second number: ");
        output.println(scanner.nextLine());
        System.out.println("Sum: " + input.readLine());
        socket.close();
    }
}

```

```

import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.PrintWriter;
import java.net.ServerSocket;
import java.net.Socket;

class Q19AdditionServer {
    public static void main(String[] args) throws Exception {
        ServerSocket serverSocket = new ServerSocket(5001);
        Socket socket = serverSocket.accept();
        BufferedReader input = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
        PrintWriter output = new PrintWriter(socket.getOutputStream(), true);
        int firstNumber = Integer.parseInt(input.readLine());
        int secondNumber = Integer.parseInt(input.readLine());
        output.println(firstNumber + secondNumber);
        socket.close();
        serverSocket.close();
    }
}

```

```

PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin\java.exe' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspace\hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q19AdditionClient'
Enter first number: 20
Enter second number: 40
Sum: 60
PS D:\Sem-5\AJP\PBL_1>

```

10. Develop a TCP Client-Server calculator application.

```

import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.PrintWriter;
import java.net.Socket;
import java.util.Scanner;

class Q20CalculatorClient {
    public static void main(String[] args) throws Exception {
        Socket socket = new Socket("localhost", 5002);
        BufferedReader input = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
        PrintWriter output = new PrintWriter(socket.getOutputStream(), true);
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter first number: ");
        output.println(scanner.nextLine());
        System.out.print("Enter operator (+, -, *, /): ");
        output.println(scanner.nextLine());
        System.out.print("Enter second number: ");
        output.println(scanner.nextLine());
        System.out.println("Result: " + input.readLine());
        socket.close();
    }
}

```

```

import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.PrintWriter;
import java.net.ServerSocket;
import java.net.Socket;

class Q20CalculatorServer {
    public static void main(String[] args) throws Exception {
        ServerSocket serverSocket = new ServerSocket(5002);
        Socket socket = serverSocket.accept();
        BufferedReader input = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
        PrintWriter output = new PrintWriter(socket.getOutputStream(), true);
    }
}

```

```

        double firstNumber = Double.parseDouble(input.readLine());
        String operator = input.readLine();
        double secondNumber = Double.parseDouble(input.readLine());
        String result;
        switch (operator) {
            case "+":
                result = String.valueOf(firstNumber + secondNumber);
                break;
            case "-":
                result = String.valueOf(firstNumber - secondNumber);
                break;
            case "*":
                result = String.valueOf(firstNumber * secondNumber);
                break;
            case "/":
                result = String.valueOf(firstNumber / secondNumber);
                break;
            default:
                result = "Invalid operator";
        }
        output.println(result);
        socket.close();
        serverSocket.close();
    }
}

```

```

● PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin
  ages' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\worksp
  hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q20CalculatorClient'
  Enter first number: 19
  Enter operator (+, -, *, /): *
  Enter second number: 2
  Result: 38.0
○ PS D:\Sem-5\AJP\PBL_1> 

```

11. Write a UDP Client using DatagramSocket.

```

import java.net.DatagramPacket;
import java.net.DatagramSocket;
import java.net.InetAddress;
import java.util.Scanner;

class Q21UDPCClient {
    public static void main(String[] args) throws Exception {
        DatagramSocket socket = new DatagramSocket();
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter message: ");
        String message = scanner.nextLine();
    }
}

```

```

        byte[] data = message.getBytes();
        InetAddress address = InetAddress.getByName("localhost");
        DatagramPacket packet = new DatagramPacket(data, data.length, address, 5003);
        socket.send(packet);
        byte[] buffer = new byte[1024];
        DatagramPacket response = new DatagramPacket(buffer, buffer.length);
        socket.receive(response);
        System.out.println("Server: " + new String(response.getData(), 0,
response.getLength()));
        socket.close();
    }
}

```

```

● PS D:\Sem-5\AJP\PBL_1> d:; cd 'd:\Sem-5\AJP\PBL_1'; & 'C:\Program Files\Java\jdk-11.0.2\bin\java.exe' -cp 'C:\Users\BAPS\AppData\Local\Temp\27c2fddc87b1f215e91539285\redhat.java\jdt_ws\PBL_1_43a4a77f\bin'
Enter message: hello
Server: Received: hello
○ PS D:\Sem-5\AJP\PBL_1>

```

12. Write a UDP Server using DatagramSocket.

```

import java.net.DatagramPacket;
import java.net.DatagramSocket;

class Q22UDPServer {
    public static void main(String[] args) throws Exception {
        DatagramSocket socket = new DatagramSocket(5003);
        byte[] buffer = new byte[1024];
        DatagramPacket packet = new DatagramPacket(buffer, buffer.length);
        socket.receive(packet);
        String message = new String(packet.getData(), 0, packet.getLength());
        System.out.println("Client: " + message);
        String reply = "Received: " + message;
        byte[] data = reply.getBytes();
        DatagramPacket response = new DatagramPacket(data, data.length,
packet.getAddress(), packet.getPort());
        socket.send(response);
        socket.close();
    }
}

```

```

● PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin\java.
  ages' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspaceSto
  hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q22UDPServer'
  Client: hello
○ PS D:\Sem-5\AJP\PBL_1>

```

13. Transfer a text file from Client to Server using TCP Socket Programming.

```

import java.io.DataOutputStream;
import java.io.File;
import java.io.FileInputStream;
import java.net.Socket;
import java.util.Scanner;

class Q23FileTransferClient {
    public static void main(String[] args) throws Exception {
        Socket socket = new Socket("localhost", 5004);
        DataOutputStream output = new DataOutputStream(socket.getOutputStream());
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter text file path: ");
        File file = new File(scanner.nextLine());
        FileInputStream fileInputStream = new FileInputStream(file);
        output.writeUTF(file.getName());
        output.writeLong(file.length());
        byte[] buffer = new byte[4096];
        int bytesRead;
        while ((bytesRead = fileInputStream.read(buffer)) != -1) {
            output.write(buffer, 0, bytesRead);
        }
        fileInputStream.close();
        output.close();
        socket.close();
    }
}

```

```

import java.io.DataInputStream;
import java.io.FileOutputStream;
import java.net.ServerSocket;
import java.net.Socket;

class Q23FileTransferServer {
    public static void main(String[] args) throws Exception {
        ServerSocket serverSocket = new ServerSocket(5004);
        Socket socket = serverSocket.accept();
        DataInputStream input = new DataInputStream(socket.getInputStream());
        String fileName = input.readUTF();
    }
}

```

```

        long size = input.readLong();
        FileOutputStream fileOutputStream = new FileOutputStream("received_" +
fileName);
        byte[] buffer = new byte[4096];
        long remaining = size;
        while (remaining > 0) {
            int bytesRead = input.read(buffer, 0, (int) Math.min(buffer.length,
remaining));
            if (bytesRead == -1) {
                break;
            }
            fileOutputStream.write(buffer, 0, bytesRead);
            remaining -= bytesRead;
        }
        fileOutputStream.close();
        socket.close();
        serverSocket.close();
        System.out.println("File received");
    }
}

```

```

● PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin\java
ages' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspaceSt
hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q23FileTransferClient'
Enter text file path: D:\Sem-5\AJP\PBL_1\practicals\data.txt
○ PS D:\Sem-5\AJP\PBL_1>

```

```

● PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin\java
ages' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspaceSt
hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q23FileTransferServer'
File received
○ PS D:\Sem-5\AJP\PBL_1>

```

14. Implement a simple Chat Application using Socket Programming.

```

import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.PrintWriter;
import java.net.Socket;
import java.util.Scanner;

class Q24ChatClient {
    public static void main(String[] args) throws Exception {
        Socket socket = new Socket("localhost", 5005);
        BufferedReader input = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
        PrintWriter output = new PrintWriter(socket.getOutputStream(), true);
        Scanner scanner = new Scanner(System.in);
    }
}

```

```

        while (true) {
            System.out.print("Client: ");
            String message = scanner.nextLine();
            output.println(message);
            if (message.equalsIgnoreCase("bye")) {
                break;
            }
            String reply = input.readLine();
            if (reply == null) {
                break;
            }
            System.out.println("Server: " + reply);
            if (reply.equalsIgnoreCase("bye")) {
                break;
            }
        }
        socket.close();
    }
}

```

```

import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.PrintWriter;
import java.net.ServerSocket;
import java.net.Socket;
import java.util.Scanner;

class Q24ChatServer {
    public static void main(String[] args) throws Exception {
        ServerSocket serverSocket = new ServerSocket(5005);
        Socket socket = serverSocket.accept();
        BufferedReader input = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
        PrintWriter output = new PrintWriter(socket.getOutputStream(), true);
        Scanner scanner = new Scanner(System.in);
        while (true) {
            String message = input.readLine();
            if (message == null) {
                break;
            }
            System.out.println("Client: " + message);
            if (message.equalsIgnoreCase("bye")) {
                break;
            }
        }
        System.out.print("Server: ");
        String reply = scanner.nextLine();
        output.println(reply);
        if (reply.equalsIgnoreCase("bye")) {

```

```

        break;
    }
}
socket.close();
serverSocket.close();
}
}

```

```

PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin\jav
ages' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspacesS
hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q24ChatClient'
Client: hello
Server: hii
Client: how are you?
Server: I am fine.
Client: 

```

15. Create a Client-Server application where the client sends student name and enrollment number and the server displays them.

```

import java.io.DataOutputStream;
import java.net.Socket;
import java.util.Scanner;

class Q25StudentClient {
    public static void main(String[] args) throws Exception {
        Socket socket = new Socket("localhost", 5006);
        DataOutputStream output = new DataOutputStream(socket.getOutputStream());
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter student name: ");
        output.writeUTF(scanner.nextLine());
        System.out.print("Enter enrollment number: ");
        output.writeUTF(scanner.nextLine());
        output.close();
        socket.close();
    }
}

```

```

import java.io.DataInputStream;
import java.net.ServerSocket;
import java.net.Socket;

class Q25StudentServer {
    public static void main(String[] args) throws Exception {
        ServerSocket serverSocket = new ServerSocket(5006);
        Socket socket = serverSocket.accept();
    }
}

```

```

        DataInputStream input = new DataInputStream(socket.getInputStream());
        String studentName = input.readUTF();
        String enrollmentNumber = input.readUTF();
        System.out.println("Student Name: " + studentName);
        System.out.println("Enrollment Number: " + enrollmentNumber);
        socket.close();
        serverSocket.close();
    }
}

```

```

● PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin\jav
ages' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspaceSt
hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q25StudentClient'
Enter student name: Dhwanit Patel
Enter enrollment number: 240170132044
○ PS D:\Sem-5\AJP\PBL_1> 

```

```

● PS D:\Sem-5\AJP\PBL_1> & 'C:\Program Files\Java\jdk-25\bin\java
ages' '-cp' 'C:\Users\BAPS\AppData\Roaming\Code\User\workspaceSt
hat.java\jdt_ws\PBL_1_43a4a77f\bin' 'Q25StudentServer'
Student Name: Dhwanit Patel
Enrollment Number: 240170132044
○ PS D:\Sem-5\AJP\PBL_1> 

```